



APPLICATION & TECHNICAL REFERENCE

Ply Legal Client Portal

Features, architecture, APIs, data storage, configuration, and codebase guide

Product UI: Ply Legal | Client Portal

Repository: [validifypro-visa-portal](#)

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Documentation basis: This guide was produced from the working repository and configuration files, not from the historical planning documents. Where a dependency or route exists but is not active, it is explicitly labelled as legacy, optional, or a stub.

Document conventions

- “Firestore” means the default Cloud Firestore database in region asia-east2.
- “Matter” means a Zoho CRM Deal linked to a portal application through zohold.
- Paths such as applications/{appId}/data/questionnaire are Firestore document paths.
- “Route-level auth” describes checks implemented by the Next.js route handler itself.

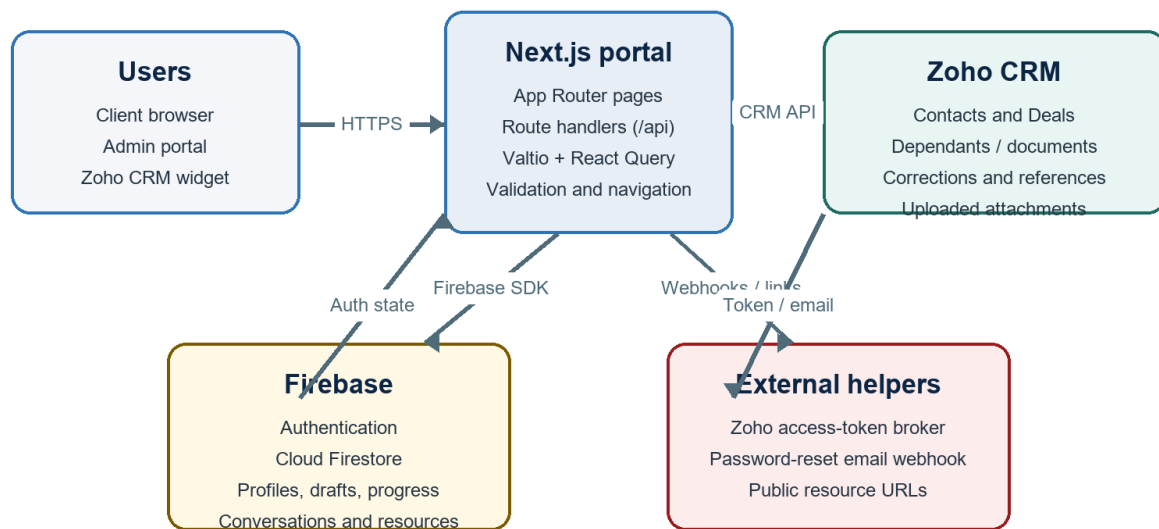
1. Executive overview

The Ply Legal Client Portal is a responsive immigration case-management portal. It lets clients sign in, maintain a profile, view visa matters sourced from Zoho CRM, complete long multi-step questionnaires, upload requested documents, review resources, submit corrections, and exchange case messages. Staff can view client conversations through a small admin surface or a Zoho CRM widget.

Primary design: Next.js provides the web interface and server routes. Firebase provides identity and the portal's working data. Zoho CRM remains the business system of record for contacts, matters, dependant records, requested documents, uploaded files, corrections, and CRM references.

Area	Current implementation
Frontend	Next.js App Router with React; 131 page routes
Server API	34 Next.js route-handler files under app/api
Identity	Firebase Authentication, email/password, persistent browser sessions
Portal database	Cloud Firestore, default database in asia-east2
CRM	Zoho CRM REST APIs; Australian data centre by default
Uploads	Zoho Matter_Documents records and their Attachments
Questionnaires	110 intake pages across partner, protection, and temporary-work flows
Deployment	Replit autoscale configuration; Next.js listens on port 5000

Current application architecture



Files are uploaded to Zoho Matter_Documents attachments; Firebase Storage is not used.

Figure 1. Current runtime architecture and data ownership.

2. Users and product scope

2.1 Client applicant

- Signs in only when the email exists in Zoho Contacts and Portal_Access is Active.
- Completes profile, address, dependant, and visa-questionnaire information.
- Views all linked visa applications and their completion progress.
- Uploads evidence against Zoho Matter Document requests.
- Reads shared or visa-specific resources, submits document corrections, and exchanges messages.

2.2 Portal administrator or case staff

- Uses the admin conversation list and conversation detail pages when the Firestore user role is admin.
- Can reply from the portal or from the Zoho CRM widget through X-Admin-Key protected endpoints.
- Receives client message references and unread flags in the Zoho Client_Messages module.

2.3 Zoho workflow integration

- Can provision or disable Firebase users through webhook-style endpoints.
- Pushes contact updates into Firestore and receives profile, dependant, questionnaire-status, document, correction, and message updates.
- Uses a separate access-token broker URL; OAuth refresh credentials are not stored in this codebase.

2.4 Public review recipient

A standalone route at `/review-pdf/{matterId}` renders questionnaire responses and supports browser print-to-PDF. It does not require a Firebase session; Firestore access therefore depends on the application's `publicReviewAccess` flag.

3. Feature catalogue

Feature	What it provides
Access and sessions	Zoho eligibility gate, Firebase email/password login, browser-local persistence, sign-out, access-denied handling.
Password lifecycle	Forced first-login password change, forgot-password token generation, reset webhook, token hashing, Firebase token revocation.

Feature	What it provides
Profile management	Personal details, address, contact data, dependants, validation, fetch from Zoho, and two-way Zoho sync.
Application dashboard	Loads Firestore applications, refreshes linked Zoho Deals, maps visa types/statuses, removes duplicates/stale records.
Questionnaire workspace	Progress dashboard, ordered intake navigation, save draft, continue, validation, page completion, and percentage tracking.
Applicant modelling	Main applicant, spouse/partner, children, migrating dependants, non-migrating family, sponsor, and per-profile form sections.
Cross-application reuse	Imports matched applicant and shared questionnaire sections; includes a 482-to-186 import helper.
Dynamic definitions	Code-based questionnaire definition with optional active Firestore override; currently used for the 482 character page.
Document upload	Reads Matter_Documents, validates type/size, creates or matches request records, uploads attachments, marks Awaiting Approval.
Resources	Authenticated shared resources plus visa-template notes and external links grouped by category.
Document review	Displays a document-review pane; creates and edits Zoho Corrections linked to the Matter.
Messaging	Firebase conversation history and unread flags; Zoho Client_Messages reference record; admin portal and CRM widget access.
Review and PDF	Searchable/collapsible questionnaire review with browser print output; separate matter-ID route.
Admin automation	Create/re-enable users, revoke access, receive Zoho contact updates, and count questionnaires.

3.1 Current application navigation

The contextual application sidebar currently exposes Questionnaire, Upload Documents, Resources, and Document Review. Additional implemented pages exist for messages, tasks, review, and deliverables, but they are not listed in the current contextual sidebar.

Deprecated message endpoints: `/api/messages/create`, `/api/messages/fetch`, and `/api/messages/upload-attachment` return HTTP 410. The supported messaging endpoints are `/api/chat/messages` and `/api/chat/send`; message attachments are not implemented.

4. Supported visa and intake flows

Public slug	Internal flow	Pages	Scope
186	temporary-work	Shared 38-page implementation	Employer Nomination Scheme; rewrite injects <code>__subclass=186</code> .

Public slug	Internal flow	Pages	Scope
482	temporary-work	Shared 38-page implementation	Skills in Demand; rewrite injects __subclass=482.
820 / partner	partner	43 pages	Partner, spouse, children, sponsor, relationship, and all-applicant sections.
866 / protection	protection	29 pages	Protection applicant, spouse, children, employment, relationship, and shared sections.

4.1 Temporary-work questionnaire

- Getting started and included-applicant selection.
- Main applicant: details, other names, identity, employment, education, language, skills, and contact details.
- Spouse/partner and child sections, including identity, education, language, custody, and other details.
- Migrating dependants and non-migrating family members.
- All applicants: addresses, countries of residence, contact details, visas, travel history, health, and character.
- Review/submit with strict completion checks and Zoho questionnaire-status update.

4.2 Partner questionnaire

- Main applicant, spouse/partner, children, and included dependants.
- Family sponsor: details, identity, family, circumstances, addresses, contacts, previous sponsorship, travel, and character.
- Relationship evidence: current relationship, relationship details, previous relationships, and supporting witnesses.
- All applicants: addresses, contacts, visas, travel, future travel/address, health, character, and personal contacts.

4.3 Protection questionnaire

- Main applicant, spouse/partner, children, current relationship, and employment-offer/business details.
- All applicants: addresses, contacts, visas, travel history, future travel, future addresses, health, character, and contacts.

- Review/submit uses the same shared draft and completion model as the other flows.

5. System architecture and data flow

5.1 Login and application synchronisation

1. The browser calls `/api/auth/verify-zoho` with the email address.
2. The server checks Zoho Contacts and requires `Portal_Access = Active`.
3. Firebase Authentication validates the email/password account.
4. The portal loads `users/{uid}` from Firestore and refreshes the contact's related Zoho Deals.
5. Deals are normalised into `applications/{appld}`; the Zoho Deal ID is stored as `zohold`.

5.2 Questionnaire save and submit

6. A page loads the application draft from `applications/{appld}/data/questionnaire`.
7. Forms use React Hook Form and page-specific validation; selected profile sections can auto-save after a 500 ms debounce.
8. Save operations persist first and update Valtio state after success.
9. Completion keys and the calculated percentage are stored in `applications/{appld}/data/completion`.
10. The linked Zoho Deal's `Questionnaires_Status` becomes In Progress or Submitted.

5.3 Documents and messages

- Document requirements are read from the Zoho Deal's `Matter_Documents` related list. Uploads are attached to the `Matter_Documents` record, not to Firestore.
- Messages are authoritative in Firestore `conversations/{conversationId}/messages`. Zoho receives a lightweight `Client_Messages` reference with preview, unread flags, timestamps, and portal link.
- Corrections are created and updated directly in the Zoho Corrections module and linked to the Matter.

6. Technology stack

Layer	Technology
Framework	Next.js 16.0.10, App Router, webpack builds, React Strict Mode
UI runtime	React 18.3.1 and React DOM 18.3.1

Layer	Technology
Language	Primarily JavaScript/JSX; TypeScript 5.6 compiler with allowJs=true and strict=false
Styling	Tailwind CSS 3.4, PostCSS, custom Ply Legal fonts, responsive utility classes
Component primitives	Radix UI, shadcn-style local components, Lucide icons, class-variance-authority
Forms and validation	React Hook Form, Zod, @hookform/resolvers
Client state	Valtio stores; TanStack React Query provider and fetch helper
Backend services	Firebase JS SDK 12.4 and Firebase Admin SDK 13.5
CRM integration	Axios, form-data, and a custom ZohoCRMClient
Identifiers and dates	nanoid and date-fns
Testing	Playwright 1.60 plus small Node-style unit tests
Package and runtime	pnpm lockfile; application targets Node.js 20+, Firebase Functions declare Node.js 22

6.1 Installed but not active in the primary runtime

- PostgreSQL/Neon/Drizzle: dependencies and drizzle.config.ts exist, but the Postgres adapter is a TODO stub and shared/schema.ts is absent.
- Firebase Functions: functions/index.js contains the starter template and exports no active business function.
- Express, Passport, WebSocket, Vite, and Wouter packages remain installed from an earlier architecture but are not the current Next.js application path.
- Firebase Storage is configured, but storage.rules denies every read and write.

7. External APIs and integrations

Integration	Endpoint/configuration	Use
Zoho CRM REST API v7	https://www.zohoapis.{datacenter}/crm/v7	Contacts, Deals, related lists, CRUD, search, and COQL. Default datacenter is com.au.
Zoho attachment API v8	https://www.zohoapis.{datacenter}/crm/v8	Uploads files to Matter_Documents/{id}/Attachments.
Zoho token broker	ACCESSTOKEN_URL or ZOHO_ACCESS_TOKEN_URL	Returns an OAuth access token; cached in server memory for 50 minutes.
Firebase Authentication	Firebase JS/Admin SDK	Email/password sign-in, account provisioning, password changes, disabling users, and token revocation.
Cloud Firestore	Firebase JS/Admin SDK	Portal profiles, applications, questionnaire drafts, completion, messages, resources, and definitions.

Integration	Endpoint/configuration	Use
Firestore REST API	firestore.googleapis.com/v1	The Zoho Deal refresh route queries and writes application documents using the caller's ID token.
Password-reset email webhook	PORTAL_RESET_EMAIL_WEBHOOK_URL	Receives email, resetLink, and contactId. If absent, the link is generated but not emailed.
External resources	Stored URLs, including Zoho WorkDrive links	The Resources page opens active external links; the portal does not copy those files.
Brand asset CDN	cdn.prod.website-files.com	Provides the favicon declared by the root layout.

7.1 Zoho modules used

Module	Portal usage
Contacts	Portal eligibility, profile fields, password flags/reset token, owner/account metadata.
Deals	Visa matters/applications, Stage, visa type, closing date, questionnaire status, documents_json.
Partner_Dependents	Spouse, children, other dependants, and non-migrating family members.
Matter_Documents	Requested-document records, status, comments, serial order, and file attachments.
Corrections	Field-level review issues linked to a Matter.
Client_Messages	Conversation reference, latest preview/time, unread flags, Firebase conversation ID, and portal link.

Not used: The codebase contains no payment, mapping, analytics, generative-AI, or SMS API integration.

8. Internal API route catalogue

All routes are implemented as Next.js App Router route handlers. “None” below means the handler does not verify a Firebase token, admin role, or shared key itself; the absence of a route-level check should not be treated as public authorisation.

Authentication and administration

Method	Route	Purpose	Auth
POST	/api/auth/verify-zoho	Gate login by Zoho contact and Portal_Access.	None
POST	/api/auth/forgot-password	Create hashed reset token in Zoho and call email webhook.	None
POST	/api/auth/reset-password	Validate token, update Firebase password/profile, clear Zoho flags.	None
POST	/api/auth/password-changed	Clear temporary-password/reset flags in Zoho.	None
POST	/api/admin/create-user	Create or re-enable Firebase user and Firestore profile	Bearer secret if set

Method	Route	Purpose	Auth
		from Zoho.	
POST	/api/admin/revoke-access	Disable Firebase user, revoke tokens, update profile.	Bearer secret if set
POST	/api/webhooks/zoho-contact-update	Push Zoho contact/profile changes into Firestore.	Bearer secret if set

Applications, profiles, and questionnaires

Method	Route	Purpose	Auth
POST	/api/applications/fetch-zoho-deals	Fetch related Deals and reconcile Firestore applications.	ID token in body for Firestore
GET/ PATCH	/api/deals/{dealId}	Read documents_json or update Questionnaires_Status.	None
POST	/api/profile/fetch-zoho	Fetch Contact, dependants, and Deals; optionally save profile.	None
POST	/api/profile/sync-zoho	Create/update Contact and synchronise dependants.	None
GET	/api/profile/zoho-population-status	Diagnostic profile/Zoho population status.	None
GET/POST	/api/intake/dependents	Read Zoho dependants or save selected/excluded IDs.	None
POST	/api/intake/sync-dependent	Create, update, or delete Partner_Dependents records.	None
GET	/api/questionnaires/count	Count applications with meaningful questionnaire data.	Firebase bearer

Messaging

Method	Route	Purpose	Auth
GET	/api/chat/messages	Read up to 50 messages and clear reader unread flag.	Firebase bearer + ownership
POST	/api/chat/send	Write message/conversation and update Zoho reference.	Firebase bearer + ownership
GET	/api/chat/admin/conversations	List latest 100 conversations; optional unread filter.	Firebase bearer + admin role
GET	/api/chat/widget/messages	Read up to 100 messages by Zoho Deal ID.	X-Admin-Key
POST	/api/chat/widget/send	Write CRM-widget admin reply and update Zoho reference.	X-Admin-Key
POST	/api/messages/create	Deprecated; returns HTTP 410.	None
GET	/api/messages/fetch	Deprecated; returns HTTP 410.	None
POST	/api/messages/upload-attachment	Deprecated; returns HTTP 410.	None

Documents, corrections, resources, and review

Method	Route	Purpose	Auth
GET	/api/uploads/matter-documents	List/sort Deal-related Matter_Documents.	None
POST	/api/uploads/zoho	Validate and upload <=5 MB file to Zoho attachment.	None
POST	/api/uploads/matter-documents/comment	Update Matter_Documents Comments.	None
GET/ POST/ PATCH	/api/corrections	List, create, or edit Matter-linked Zoho Corrections.	None
GET	/api/resources/shared	Return active shared resources for an authorised application.	Firebase bearer + ownership
GET	/api/resources/template	Return active visa-template notes and external links.	Firebase bearer + ownership
GET	/api/review-pdf/application/{matterId}	Find a Firestore application by Zoho Matter ID.	None; Firestore rules apply
GET	/api/review-pdf/application/{matterId}/draft	Return questionnaire draft for public review.	None; Firestore rules apply

Diagnostics

Method	Route	Purpose	Auth
GET	/api/test/zoho-get	Read a Zoho Contact for integration testing.	None
POST	/api/test/zoho-update	Update a Zoho Contact for integration testing.	None
GET	/api/test/zoho-verify-token	Fetch token and verify Zoho Contacts module access.	None

9. Where information is saved

Information	Primary location	Details
User credentials	Firebase Authentication	Passwords are managed by Firebase; the portal stores no plaintext password.
Portal user profile	Firestore users/{uid}	Personal/contact data, role, access flags, Zoho IDs, sync metadata, dependant summary.
Visa application shell	Firestore applications/{appld}	Owner, type/code, status, reference, timestamps, zohold, public review flag.
Questionnaire answers	Firestore applications/{appld}/data/questionnaire	Nested answers, profiles, per-profile sections, selected dependants, visa context.
Completion/progress	Firestore applications/{appld}/data/completion	Page completion booleans and completionPercentage.
Preferences	Firestore users/{uid}/preferences/settings	Current prefill setting.
Legacy current draft	Firestore users/{uid}/drafts/current	Only used by a real-time subscription path; application-scoped

Information	Primary location	Details
		draft is primary.
Zoho contact snapshot	Firestore users/{uid}/zoho/contact	Server-synchronised Zoho data and timestamps.
Application messages	Firestore conversations/{applicationId}/messages/{messageId}	Message body, sender role/UID, and created time.
Conversation summary	Firestore conversations/{applicationId}	Client/matter references, preview, latest time, and unread flags.
Shared resources	Firestore resources/{resourceId}	Active global notes and external links.
Visa resource templates	Firestore resourceTemplates/{slug}/items/{itemId}	Active categorized notes and external links.
Dynamic questionnaire definitions	Firestore questionnaireDefinitions/{id}/pages/{pageId}	Optional active override for code-defined pages.
Uploaded client files	Zoho Matter_Documents/{id}/Attachments	PDF/JPG/PNG/DOC/DOCX/TXT, maximum 5 MB in the upload handler.
Matter and CRM records	Zoho Contacts/Deals/Partner_Dependents/Corrections/Client_Messages	CRM system-of-record entities and references.
Browser session	Firebase Auth IndexedDB persistence	Keeps the authenticated session across browser/tab closes.
Fallback browser data	localStorage adapter	Used only when configured or when adapter initialisation falls back.

9.1 Firestore path map

```

users/{uid}
├── profile and access metadata
├── preferences/settings
├── drafts/current                (legacy/current-draft listener)
└── zoho/contact                 (Zoho snapshot)

applications/{appId}
├── application metadata and zohoId
└── data/
    ├── questionnaire            (primary draft and applicant answers)
    ├── completion              (page flags and percentage)
    └── {dataType}               (adapter-supported application data)

conversations/{applicationId}
└── messages/{messageId}

resources/{resourceId}
resourceTemplates/{visaSlug}/items/{itemId}
questionnaireDefinitions/{definitionId}/pages/{pageId}

```

9.2 Browser localStorage fallback keys

```
ply_session
ply_user
ply_user_profile
intake_draft
intake_prefill
ply:applications
application-scoped draft, completion, and data keys generated by the adapter
ply-theme (ThemeProvider)
```

Important: Firebase Storage is not a document repository in this application. storage.rules denies all reads and writes. Changing upload storage requires a deliberate feature and security design, not only a Firebase configuration change.

10. Folder and code structure

```
validifypro-visa-portal/
├── app/                               Next.js App Router pages, layouts, and API handlers
│   ├── api/                          34 route-handler files
│   ├── applications/                 Application list and per-application workspace
│   ├── intake/                      110 visa-intake page files
│   │   ├── partner/                 43 pages
│   │   ├── protection/              29 pages
│   │   └── temporary-work/          38 pages shared by subclasses 186 and 482
│   ├── admin/                       Admin conversation pages
│   ├── profile/                     Profile and first-time setup
│   └── review-pdf/                   Matter-ID review and browser PDF surface
├── src/
│   └── components/                   Shared layout, navigation, form, intake, questionnaire, and UI
├── components
│   ├── hooks/                       Toast and draft-form hooks
│   └── lib/                          Firebase, Zoho, adapters, routes, validation, questionnaire
├── definitions
│   ├── stores/                      Valtio auth, applications, draft, and application-data state
│   ├── reuseable/                   Country/month option data and legacy reusable components
│   └── assets/                       Brand images
├── public/                           Fonts, logo, and login image
├── e2e/                              Playwright 186/482 questionnaire tests and harness
├── scripts/                          Seeding, backfill, and admin-page helpers
├── functions/                        Firebase Functions starter package; no active exported function
├── docs/                             Existing implementation notes and this generated guide
├── knowledgebase/                   Operational and testing notes
├── attached_assets/                 Screenshots and historical requirement material
├── firestore.json                   Firestore, Functions, and Storage configuration
├── firestore.rules                   Firestore client access rules
├── storage.rules                     Deny-all Firebase Storage rules
├── next.config.js                    Rewrites, headers, aliases, and Firebase env exposure
├── package.json                     Dependencies and scripts
└── .replit                           Node runtime, port 5000, and autoscale deployment
```

10.1 Key implementation files

File/folder	Responsibility
app/layout.js	Root metadata and provider shell.
app/providers.jsx	React Query, theme, AuthGuard, and navigation-loading providers.

File/folder	Responsibility
<code>next.config.js</code>	Visa-route rewrites, widget CORS headers, build aliases, and environment exposure.
<code>src/lib/firebase.js</code>	Browser Firebase app, Auth, Firestore, and persistent login.
<code>src/lib/firebase-admin.js</code>	Server Firebase Admin initialisation.
<code>src/lib/serverAuth.js</code>	Firebase token verification, role checks, and widget admin-key verification.
<code>src/lib/zohoClient.js</code>	Zoho token cache and CRM CRUD/related-record/attachment helpers.
<code>src/lib/adapters/firebase.js</code>	Current persistence adapter and application/draft/profile operations.
<code>src/stores/draftStore.js</code>	Questionnaire state, save/completion/import, applicant management, and CRM sync.
<code>src/lib/routes.js</code>	Visa-specific navigation, rewrites, and completion paths.
<code>src/lib/questionnaires/</code>	Definition-based questionnaire loading, validation, and Firestore override.
<code>app/api/</code>	Server routes integrating Firebase, Zoho, resources, messaging, and uploads.
<code>firebase.rules</code>	Owner-based client access and public-review exception.
<code>e2e/</code>	Playwright end-to-end questionnaire coverage.

10.2 Route aliases

Public application route	Internal destination
<code>/applications/186/{appId}/intake/*</code>	<code>/intake/temporary-work/*?applicationId={appId}&__subclass=186</code>
<code>/applications/482/{appId}/intake/*</code>	<code>/intake/temporary-work/*?applicationId={appId}&__subclass=482</code>
<code>/applications/820/{appId}/intake/*</code>	<code>/intake/partner/*?applicationId={appId}</code>
<code>/applications/866/{appId}/intake/*</code>	<code>/intake/protection/*?applicationId={appId}</code>
<code>/applications/partner/{appId}/intake/*</code>	<code>/intake/partner/*?applicationId={appId}</code>
<code>/applications/protection/{appId}/intake/*</code>	<code>/intake/protection/*?applicationId={appId}</code>

11. Configuration and environment variables

Variable	Use	Notes
<code>NEXT_PUBLIC_DATABASE_TYPE</code>	Required	Set to firebase in the current environment. Other accepted values: <code>localStorage</code> , <code>postgres</code> .
<code>NEXT_PUBLIC_FIREBASE_API_KEY</code>	Required	Firebase web configuration.
<code>NEXT_PUBLIC_FIREBASE_AUTH_DOMAIN</code>	Required	Firebase web configuration.
<code>NEXT_PUBLIC_FIREBASE_PROJECT_ID</code>	Required	Firebase web/Admin project and Firestore REST URL.
<code>NEXT_PUBLIC_FIREBASE_STORAGE_BUCKET</code>	Required by initialiser	Configured even though Storage access is deny-all.

Variable	Use	Notes
NEXT_PUBLIC_FIREBASE_MESSAGING_SENDER_ID	Required	Firebase web configuration.
NEXT_PUBLIC_FIREBASE_APP_ID	Required	Firebase web configuration.
FIREBASE_SERVICE_ACCOUNT_KEY	Required for server routes	JSON service account supplied as a secret; used by Firebase Admin.
ACCESSTOKEN_URL	Zoho required	Preferred Zoho access-token broker URL.
ZOHO_ACCESS_TOKEN_URL	Zoho fallback	Backward-compatible token broker variable.
ZOHO_DATACENTER	Optional	Defaults to com.au.
PORTAL_ADMIN_KEY	Widget required	X-Admin-Key expected from the Zoho CRM widget.
ZOHO_WEBHOOK_SECRET	Production required	Bearer secret for user provisioning, revocation, and contact-update webhook.
PORTAL_RESET_EMAIL_WEBHOOK_URL	Password reset	Webhook that sends the generated reset link.
NEXT_PUBLIC_APP_URL / APP_URL	Optional	Base URL used to build password-reset links; localhost fallback exists.
NEXT_PUBLIC_BASE_URL	Optional	Base URL included in Zoho admin chat links.
DATABASE_URL	Inactive adapter	Required only by Drizzle tooling; Postgres adapter is not implemented.
NEXT_DIST_DIR	Optional	Overrides the Next.js build directory.
E2E_PORT / CI	Test only	Playwright port and CI behaviour.

11.1 Configuration files

File	Purpose
next.config.js	App rewrites, widget CORS headers, aliases, body-size limit, and public Firebase variables.
firebase.json	Default Firestore database in asia-east2, Storage rules, and functions codebase.
firestore.rules	Owner access, active-resource reads, admin definition/template writes, and explicit public-review reads.
storage.rules	Deny all.
tailwind.config.js/.ts	Theme, fonts, colours, and component scanning.
playwright.config.js	E2E server and browser configuration.
drizzle.config.ts	Dormant PostgreSQL migration configuration.
.replit	Node 20 module, dev/start commands, port 5000, and autoscale deployment.

12. Local development, build, and deployment

12.1 Prerequisites

- Node.js 20 or newer for the Next.js application; Node.js 22 for Firebase Functions if that package is deployed.
- pnpm is preferred because pnpm-lock.yaml is committed.
- A Firebase project with Email/Password authentication enabled, Firestore created, and the documented environment variables.
- A Zoho token-broker endpoint and CRM modules/fields matching the integration.

12.2 Common commands

```
pnpm install
pnpm dev           # Next.js dev server on port 5000
pnpm build         # production webpack build
pnpm start         # production server on port 5000
pnpm check         # TypeScript no-emit check
pnpm test:e2e      # Playwright tests
pnpm questionnaires:seed # seed active questionnaire definitions into Firestore
pnpm db:push       # Drizzle push; only relevant if Postgres work is completed
```

12.3 Deployment

The committed .replit file configures an autoscale deployment that runs npm run build and npm run start, with application port 5000 exposed externally. A Netlify Next.js plugin is installed as a development dependency, but no Netlify configuration was found. Firebase Hosting is not configured.

Firebase Functions: The functions package targets Node.js 22 and caps each function at 10 instances, but functions/index.js currently exports no deployed business handler.

13. Testing and verification

Test	Coverage
e2e/questionnaire-186.spec.js	End-to-end Employer Nomination questionnaire flow.
e2e/questionnaire-482.spec.js	End-to-end Skills in Demand questionnaire flow.
e2e/support/questionnaireHarness.js	Shared Playwright setup and questionnaire helpers.
src/lib/adapters/__tests__/factory.test.js	Adapter selection.
src/lib/adapters/__tests__/localStorage.test.js	Browser fallback persistence.
src/lib/questionnaires/__tests__/validation.node.test.js	Questionnaire-definition validation.
src/stores/__tests__/draftStore.nonMigrating.test.js	Non-migrating member draft behaviour.

Only Playwright and the TypeScript checker have package scripts. The small Node tests need an explicit runner/command if they are to be part of CI. Current E2E coverage is strongest for subclasses 186 and 482; partner, protection, messaging, uploads, webhooks, and access control need dedicated automated coverage.

14. Security and operational notes

Immediate action: The repository currently tracks `.env` and two Firebase service-account JSON files even though `.gitignore` now excludes `.env`. Treat the credentials as exposed: rotate them, remove the files from Git history, and move all secrets to the deployment secret store.

Severity	Area	Current observation	Recommended action
Critical	Tracked secrets	<code>.env</code> , <code>firebase-service-account.json</code> , and <code>validate-pro-test-firebase-adminsdk-fbsvc-4f4b7a7e17.json</code> are tracked.	Rotate credentials and purge Git history.
High	Missing route-level auth	Several profile, dependant, deal, correction, upload, review, and test routes do not verify caller identity/ownership.	Add central auth/authorisation; disable diagnostic routes in production.
High	Webhook fail-open	Webhook/admin routes enforce <code>ZOHO_WEBHOOK_SECRET</code> only when the variable is configured.	Fail closed when the secret is absent.
High	Credential logging	<code>firebase-admin.js</code> logs the length and first 50 characters of <code>FIREBASE_SERVICE_ACCOUNT_KEY</code> .	Remove all credential-derived logs.
High	Upload route	<code>/api/uploads/zoho</code> validates file type/size but does not authenticate or verify Deal ownership.	Require Firebase token and application ownership.
Medium	Public review	Matter-ID review routes have no route auth; Firestore <code>publicReviewAccess</code> is the effective gate.	Keep the flag false by default and add signed/time-limited access.
Medium	Debug ingestion	<code>draftStore</code> and one partner page call a hard-coded <code>127.0.0.1:7242</code> ingest endpoint.	Remove debug calls before release.
Medium	CORS scope	Zoho widget routes return <code>Access-Control-Allow-Origin: *</code> .	Restrict allowed origins if the widget host is stable.
Medium	Reset token lifetime	The Zoho reset-token hash has no expiry timestamp in the current flow.	Store and enforce an expiry and single-use audit data.
Low	Dormant code	Postgres, Express/Vite remnants, deprecated routes, backups, and test pages expand the maintenance surface.	Remove once confirmed unused.

14.1 Existing safeguards

- Firestore owner rules protect users and application subcollections; unrecognised paths are denied.
- Authenticated chat routes verify Firebase ID tokens and application ownership; admin conversation listing checks the admin role.
- Zoho widget routes require `PORTAL_ADMIN_KEY`.
- Login fails closed if Zoho eligibility cannot be verified.

- Password-reset responses avoid disclosing whether an email exists.
- Uploads enforce a 5 MB size limit and an allow-list of document MIME types.
- Firebase Storage denies all reads and writes.

15. Maintenance guide and appendices

15.1 Where to make common changes

Task	Primary locations
Add/edit a static intake page	app/intake/{flow}/.../page.js plus src/lib/routes.js.
Change questionnaire route order	src/lib/routes.js and completion/validation helpers.
Add a definition-driven question	src/lib/questionnaires, questionnaire renderer components, then run questionnaires:seed.
Change data persistence	src/lib/adapters/firebase.js; keep Firestore rules and server routes aligned.
Change profile/CRM mapping	src/lib/zohoClient.js and app/api/profile routes.
Change Deal/application mapping	app/api/applications/fetch-zoho-deals/route.js and src/lib/visaDisplay.js.
Change document uploads	application uploads page, /api/uploads routes, and Zoho Matter_Documents fields.
Change messaging	application/admin message pages, /api/chat routes, and Zoho Client_Messages fields.
Change resources	/api/resources routes, Firestore resource collections, and resource page.
Change access control	src/lib/serverAuth.js, each route handler, firestore.rules, and AuthGuard.
Change deployment	.replit, package scripts, runtime secrets, and Firebase/Zoho configuration.

15.2 Route surface summary

Surface	Count	Notes
Top-level page routes	131	Authentication, profile, application workspace, intake, admin, and review.
Intake pages	110	Partner 43; protection 29; temporary-work 38.
API route files	34	Auth/admin, applications, profile, intake, chat, resources, uploads, review, and tests.
Application workspace pages	9 child pages	Questionnaire, uploads, resources, corrections, messages, tasks, review, deliverables, plus layout.
Admin pages	2	Conversation list and detail.
Public review pages	1	Matter-ID questionnaire review and print.

15.3 Repository snapshot notes

- Documented commit: 86740ed, dated 2026-06-28.
- The working tree was inspected without reading or reproducing secret values.
- Historical markdown plans and attached screenshots remain useful background, but the code and active configuration were treated as authoritative.
- The product metadata names the application “Ply Legal | Client Portal”; the repository/workspace name remains validifypro-visa-portal.

End of guide: Update this document after material changes to authentication, persistence, Zoho modules, route access control, supported visa flows, or deployment.